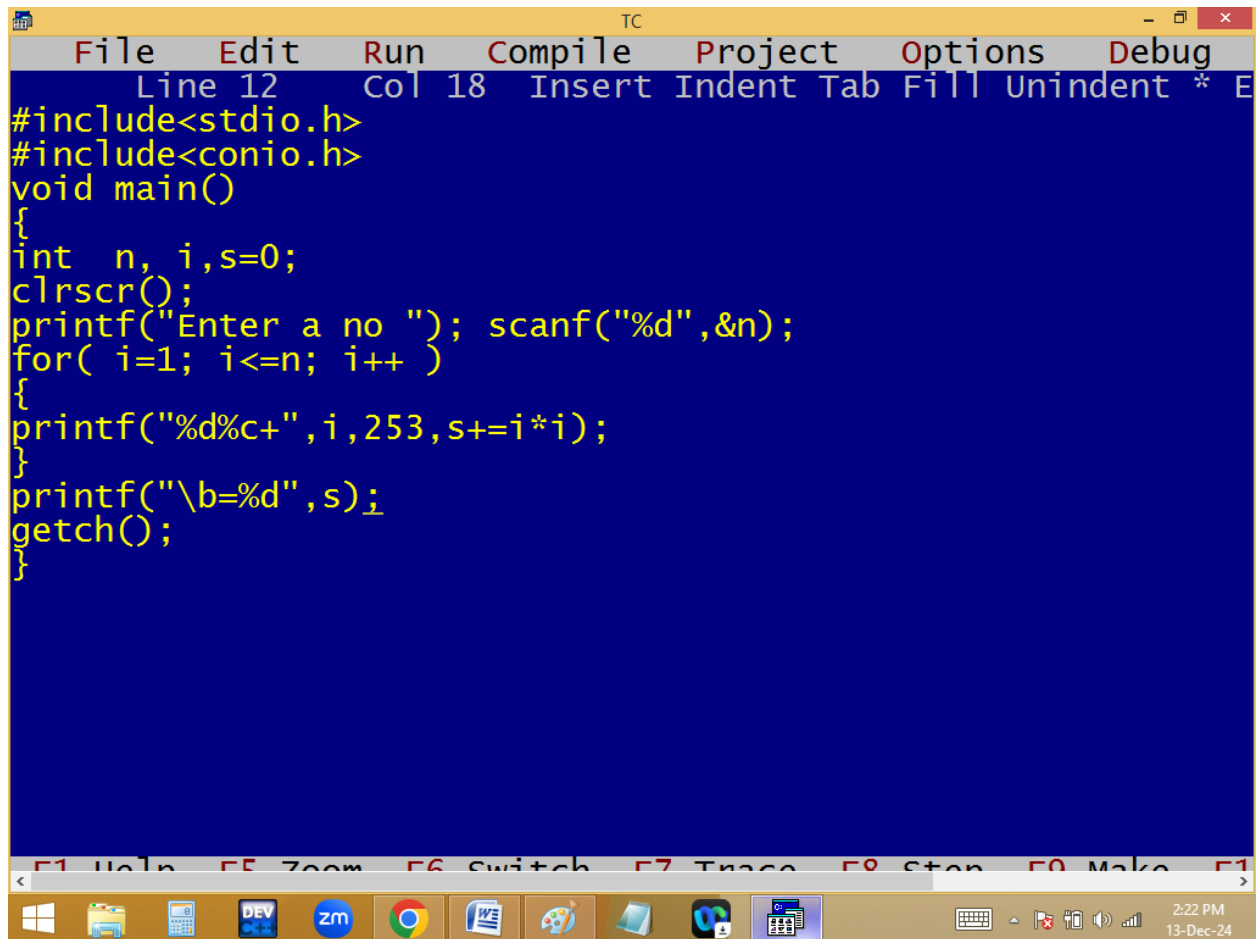


Finding square sum:

$$n=5 \rightarrow 1^2+2^2+3^2+4^2+5^2=55$$



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 12", "Col 18", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains a C program that calculates the sum of squares of the first n natural numbers. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n, i, s=0;
    clrscr();
    printf("Enter a no "); scanf("%d",&n);
    for( i=1; i<=n; i++ )
    {
        printf("%d%c+", i, 253, s+=i*i);
    }
    printf("\b=%d", s);
    getch();
}
```

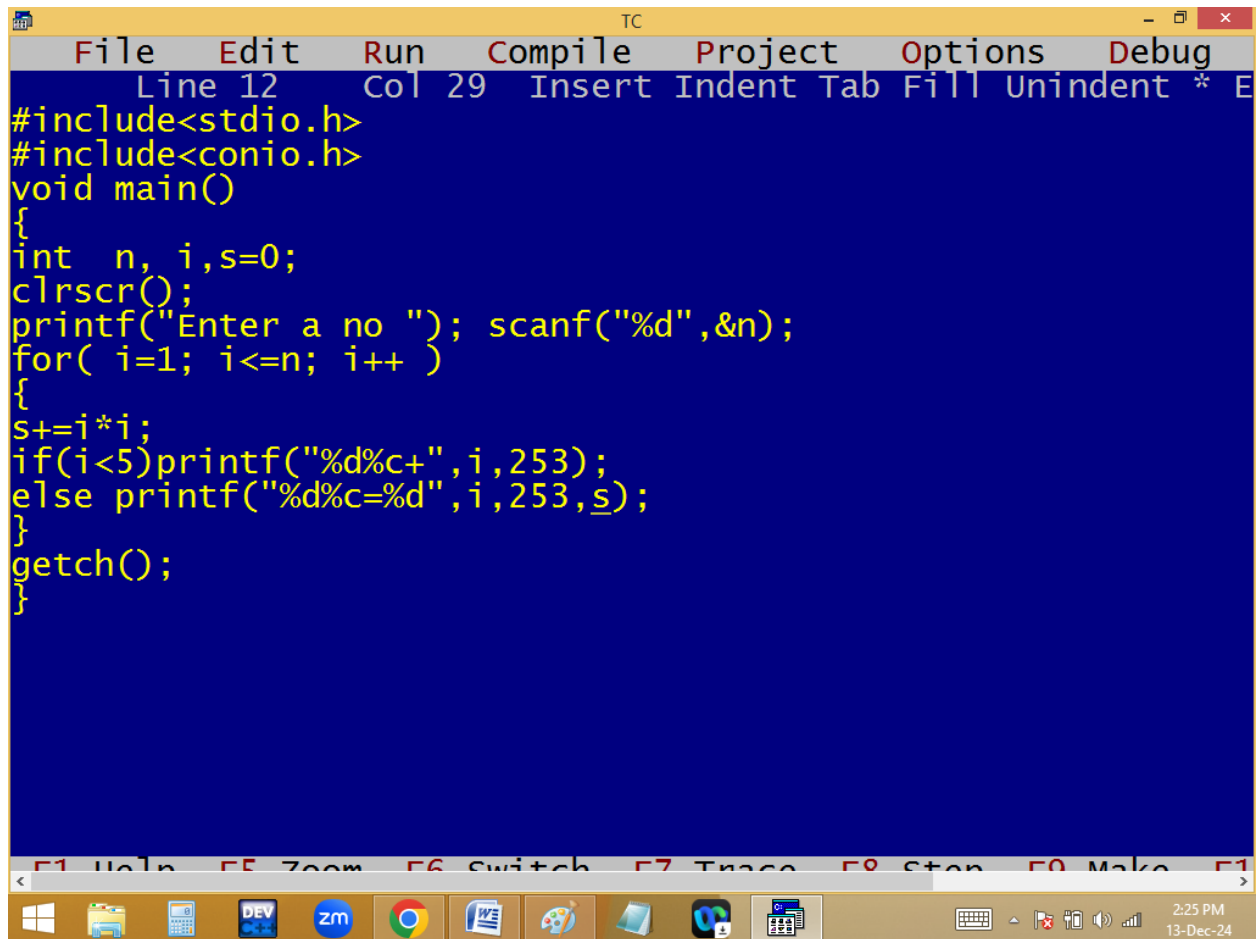
Below the code editor, there is a toolbar with icons for various functions, and a status bar at the bottom showing the time "2:22 PM" and the date "13-Dec-24".

```
TC
Enter a no 5
12+22+32+42+52=55_
```

$\frac{5}{0}$

```
for( i=1;i<=5;i++)
{
    |  ^  +
    p("%d%c+",i,253,s+=i*i);
}
p(s);
```

Without using \b:



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 29 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n, i, s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for( i=1; i<=n; i++ )
{
s+=i*i;
if(i<5)printf("%d%c+",i,253);
else printf("%d%c=%d",i,253,_s);
}
getch();
}
```

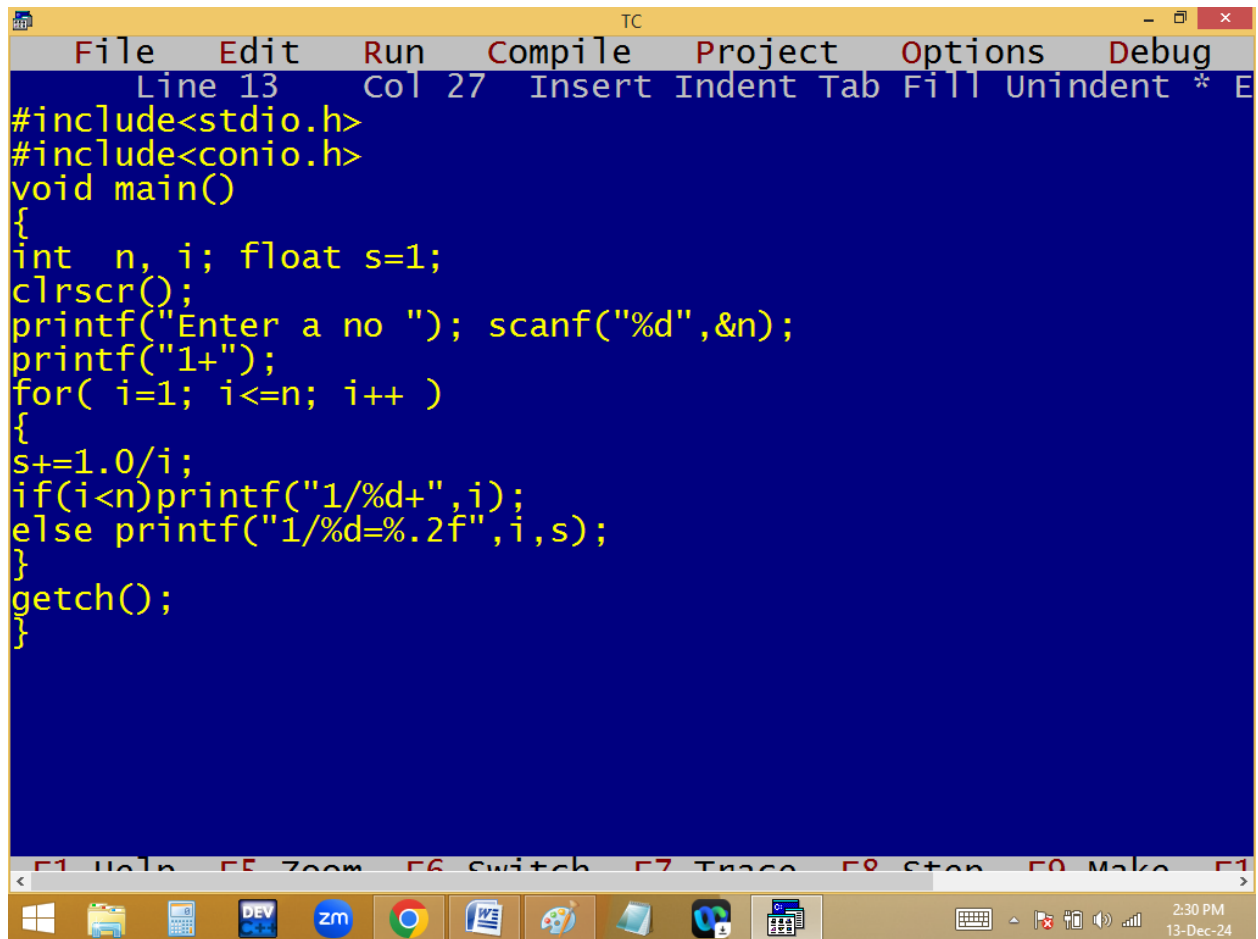
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:25 PM
13-Dec-24

```
TC
Enter a no 5
12+22+32+42+52=55_
```

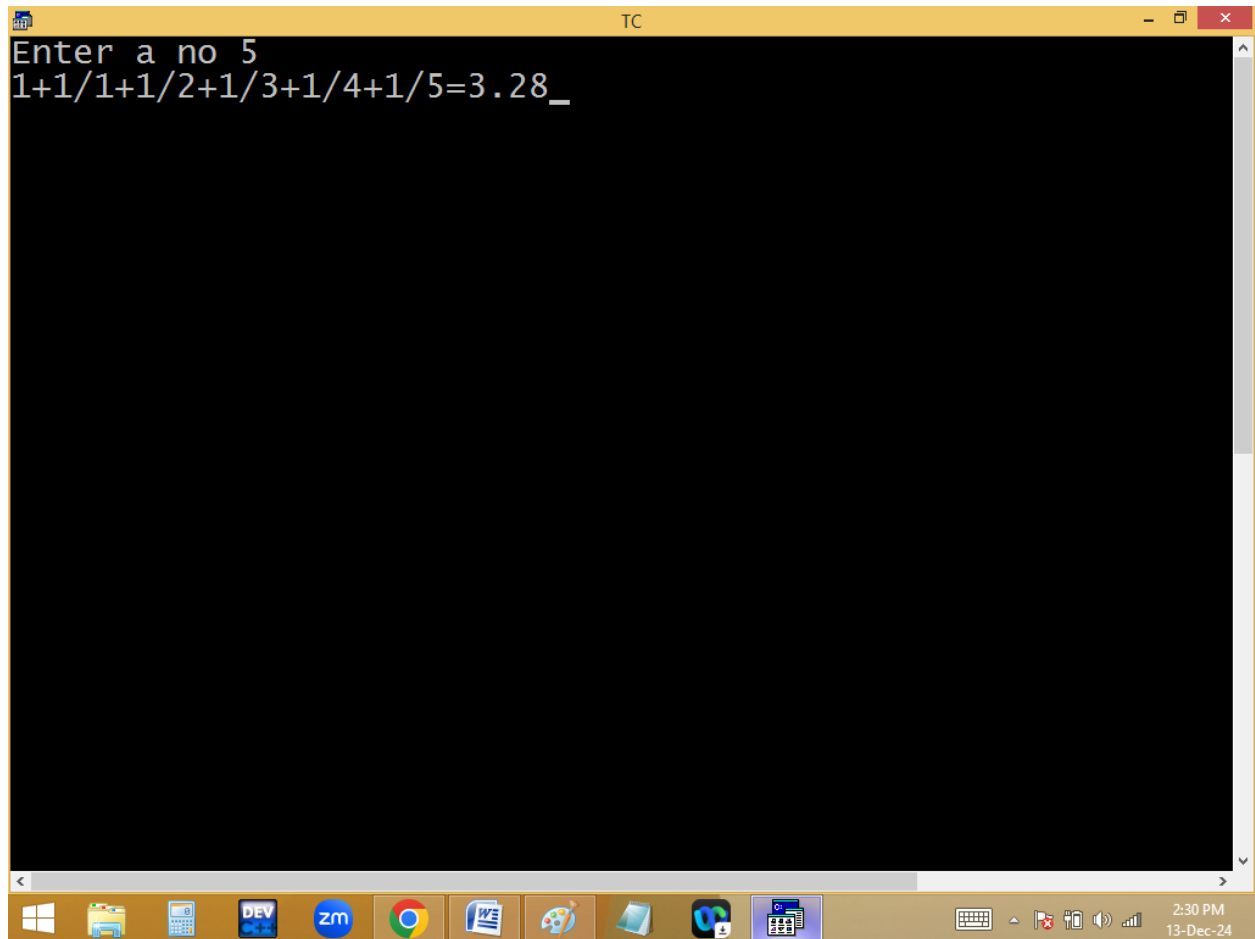
Harmonic series:

$$n=5 \rightarrow 1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = 3.28$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 13 Col 27" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The code is written in yellow text on a blue background. It includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside `main`, it declares `int n, i;` and `float s=1;`, calls `clrscr()`, prompts the user to "Enter a no ", and reads the input into `n`. A `for` loop starts at `i=1` and continues while `i<=n`. Inside the loop, `s` is updated as `s+=1.0/i;`. A conditional check `if(i<n)` prints `"1/%d+", i`, while the `else` branch prints `"1/%d=%.2f", i, s`. The program ends with `getch()` and a closing brace for `main`. The bottom status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom contains icons for various applications and the system clock, which shows 2:30 PM on 13-Dec-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n, i; float s=1;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("1+");
for( i=1; i<=n; i++ )
{
s+=1.0/i;
if(i<n)printf("1/%d+",i);
else printf("1/%d=%.2f",i,s);
}
getch();
}
```



```
Enter a no 5
1+1/1+1/2+1/3+1/4+1/5=3.28_
```

Finding Armstrong no?

1 is a single digit no $\rightarrow 1^1 = 1$

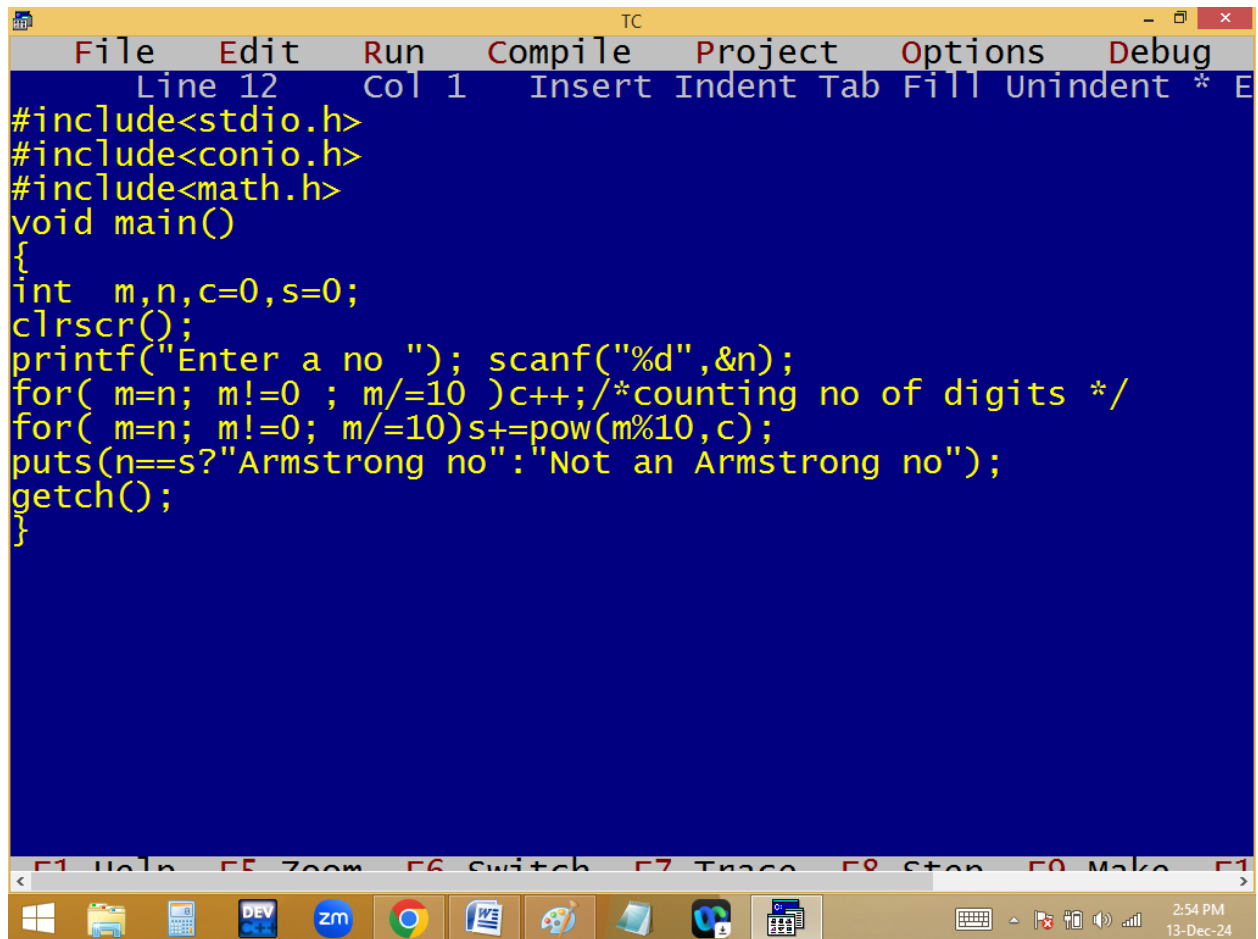
2 is a single digit no $\rightarrow 2^1 = 2$

9 is a single digit no $\rightarrow 9^1 = 9$

153 is a 3 digit no $\rightarrow 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

1634 is a 4 digit no $\rightarrow 1^4 + 6^4 + 3^4 + 4^4 = 1634$

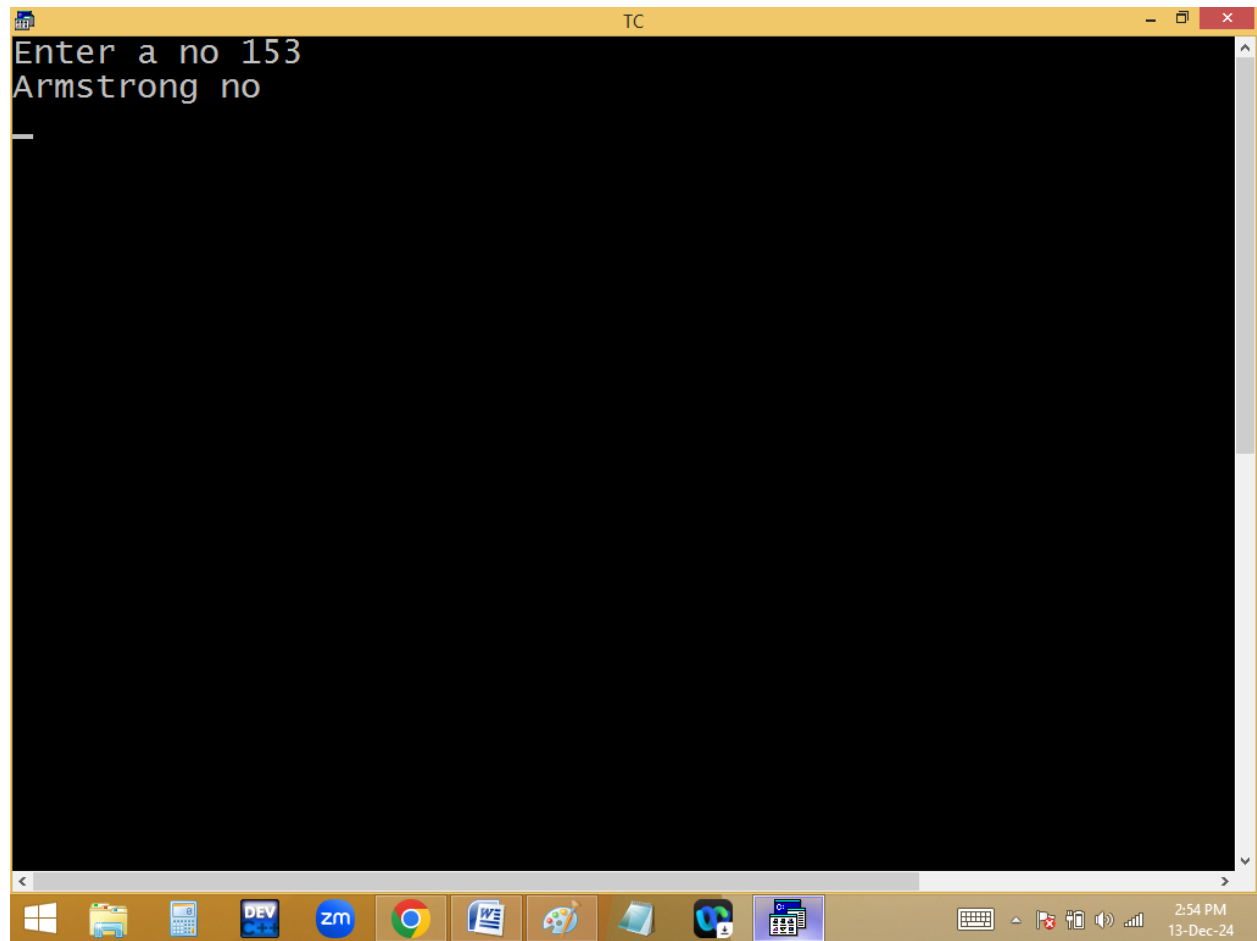
370, 371, 407, 8208,....

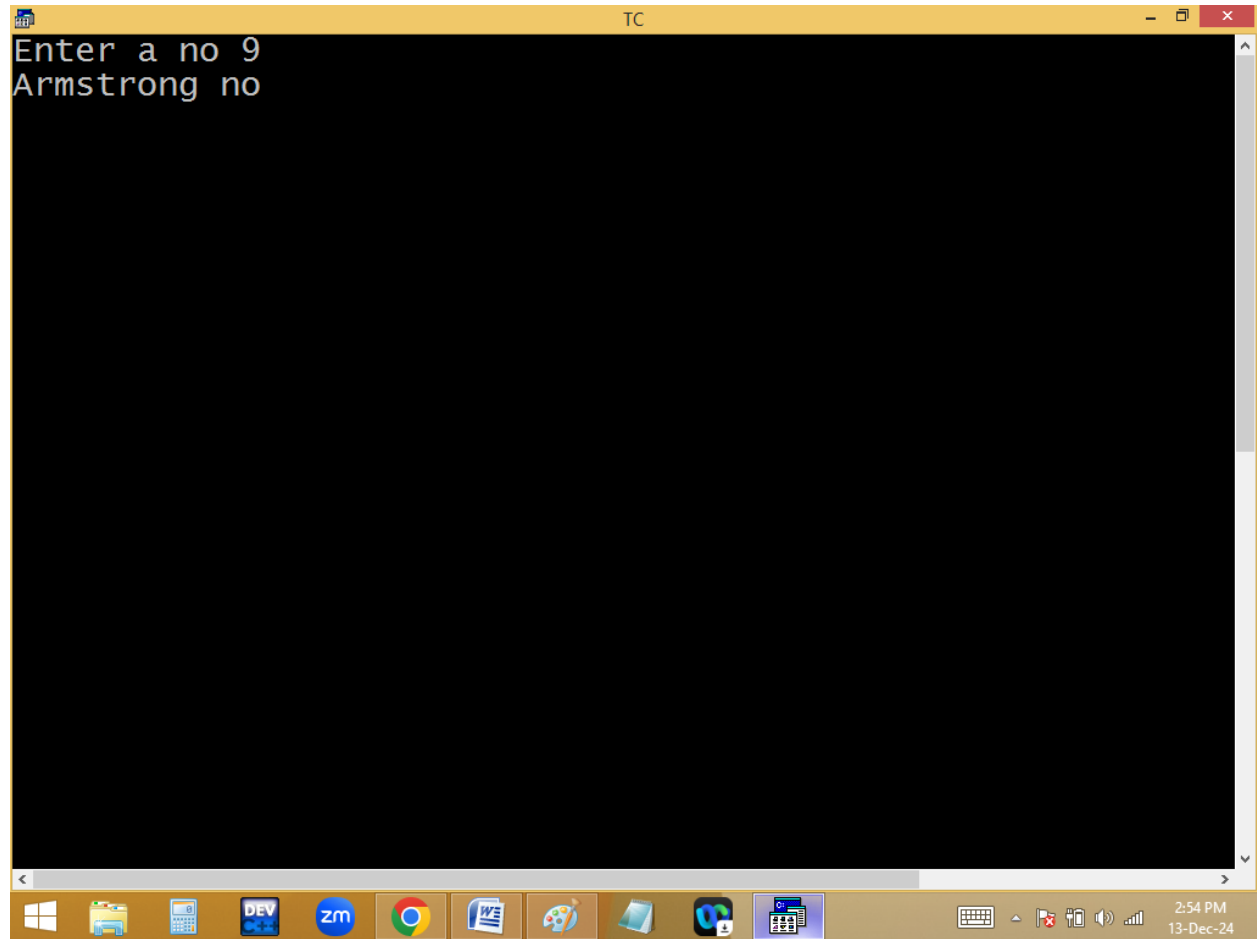


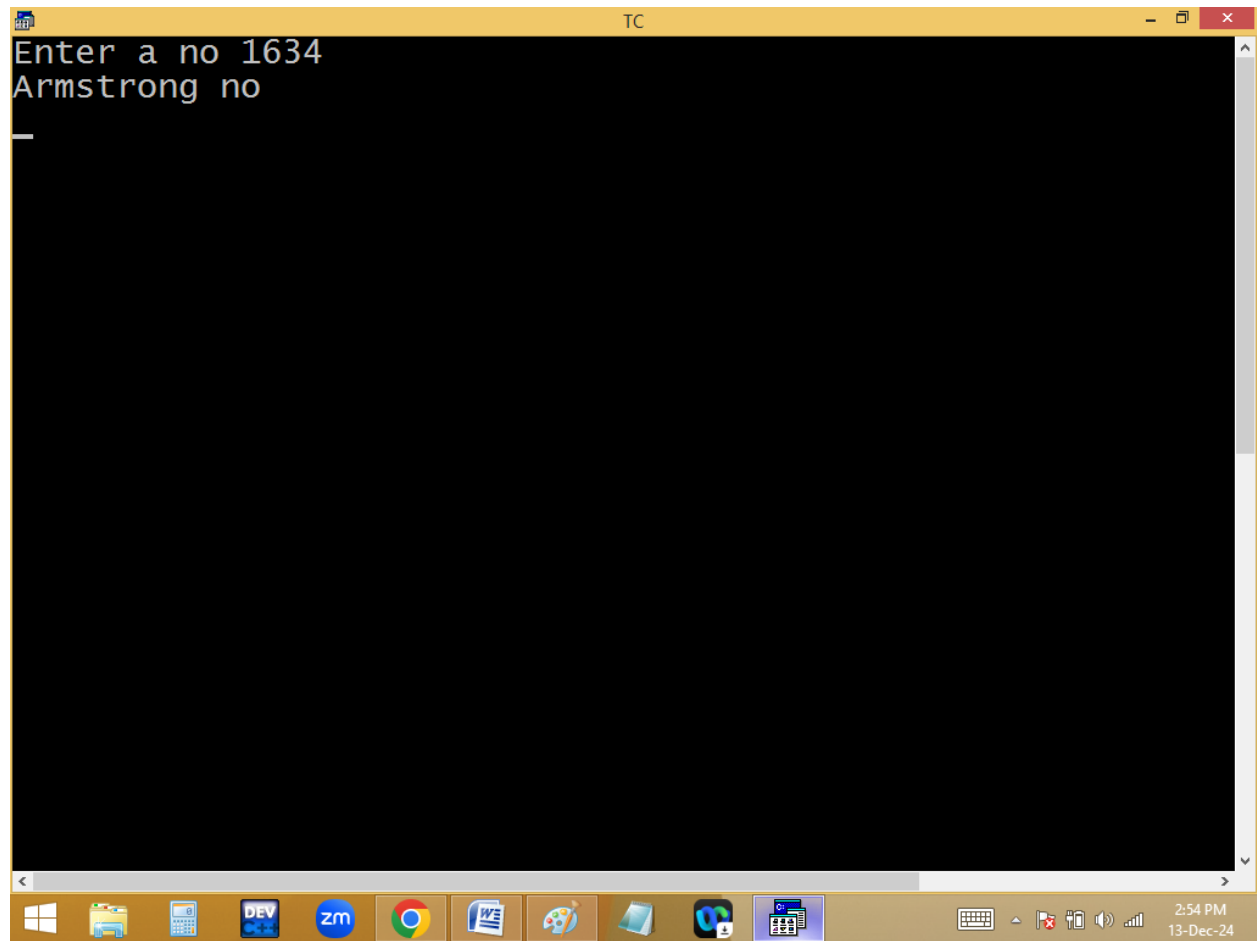
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 12", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int  m,n,c=0,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for( m=n; m!=0 ; m/=10 )c++;/*counting no of digits */
for( m=n; m!=0; m/=10)s+=pow(m%10,c);
puts(n==s?"Armstrong no":"Not an Armstrong no");
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "2:54 PM" and the date "13-Dec-24".







```
TC
Enter a no 143
Not an Armstrong no
```

for(m=n;m!=0;m/=10)c++;

1
153

m
153

c = 0 1 2 3 ✓

for(m=n; m!=0; m/=10)

s+=pow(m%10,c);

1
5
3
0

$\frac{5}{0+3^3=27}$
 $27+5^3=152$
 $152+1^3=153$

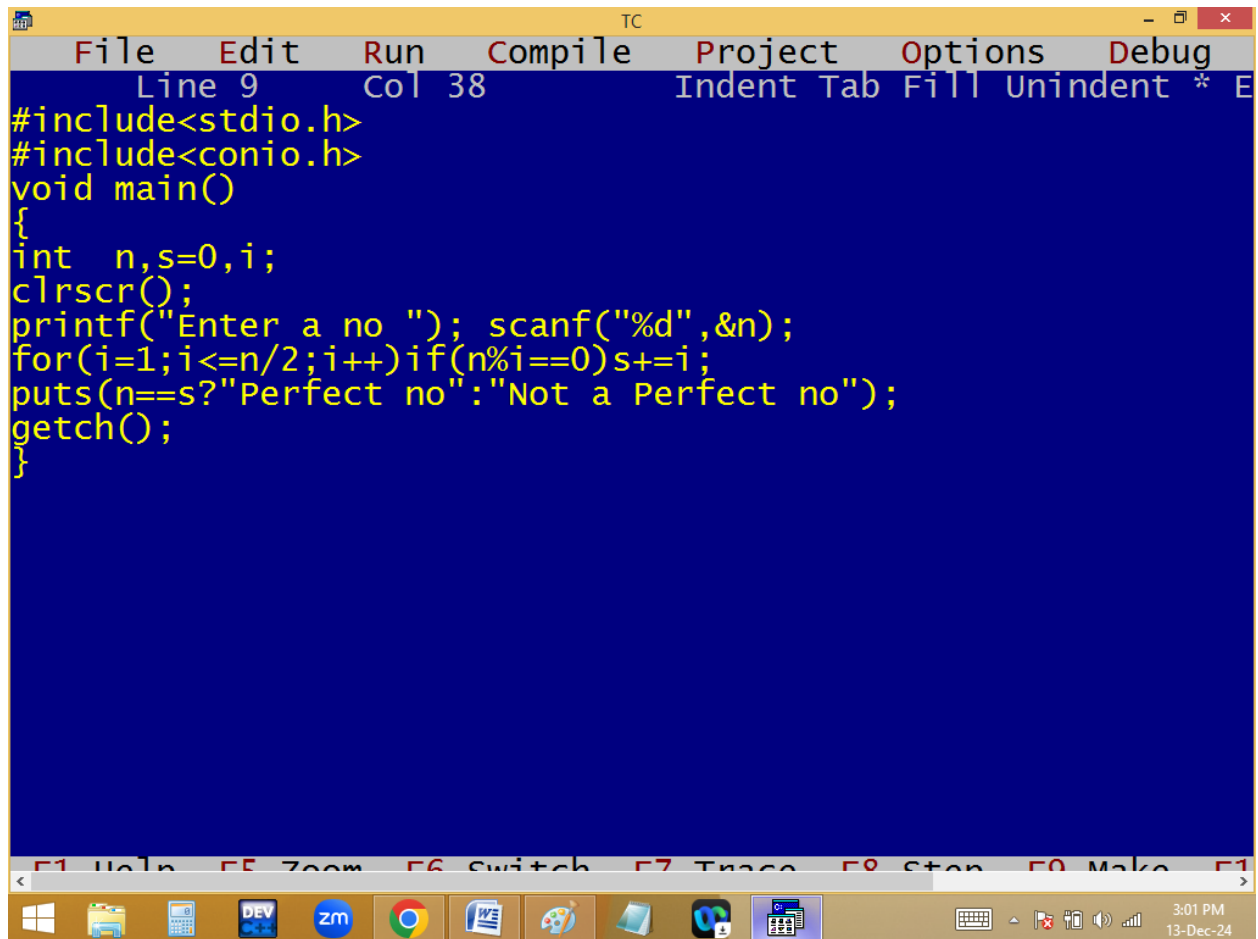
143 = 1 + 64 + 27 + 92

puts(n==s ? "Arm" : "Not");

Finding perfect no or not?

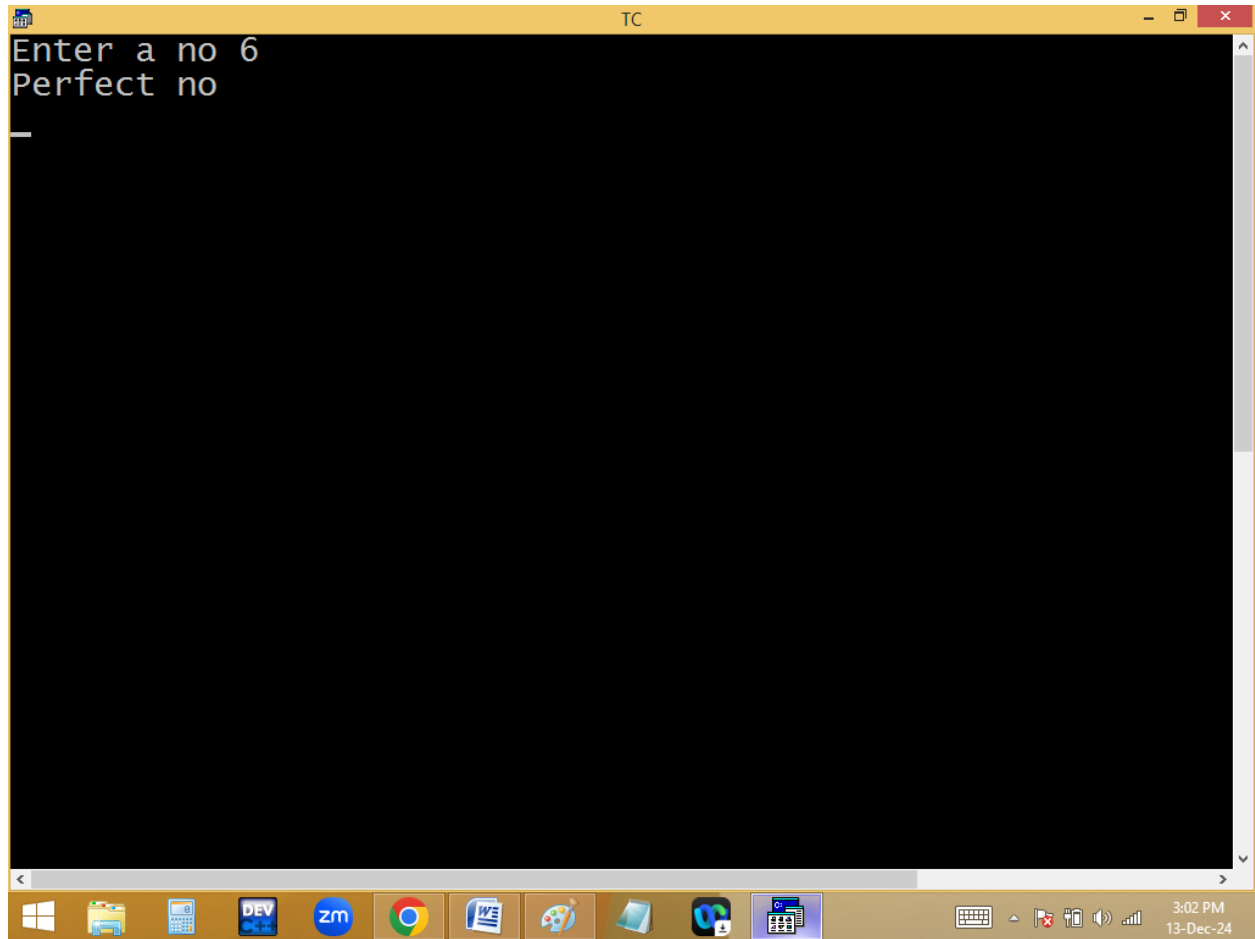
6 factors sum is $1+2+3 = 6$

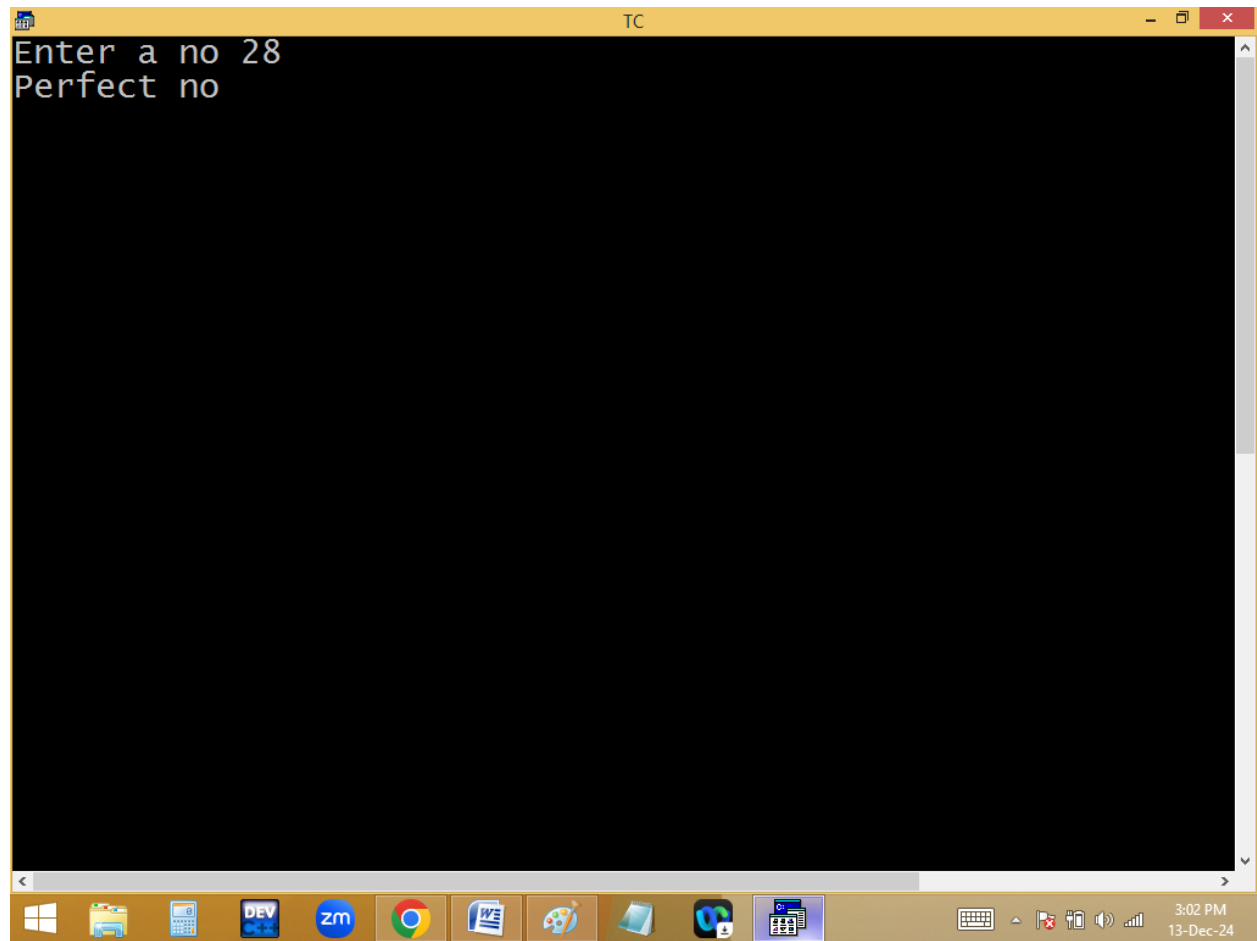
28 factors sum is $1+2+4+7+14=28$

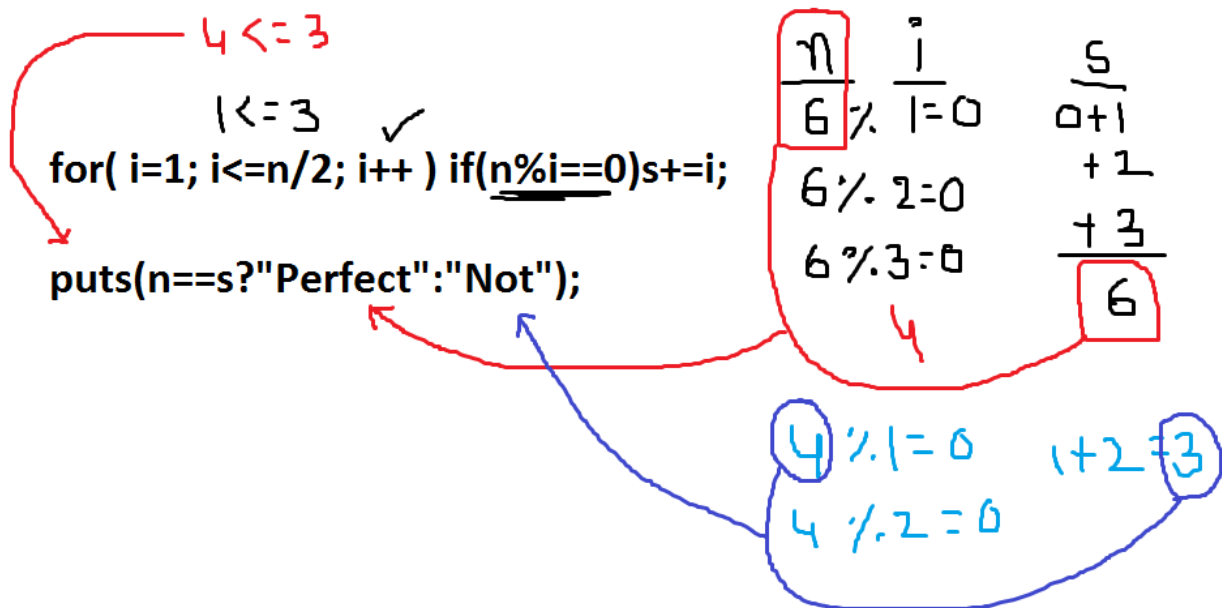
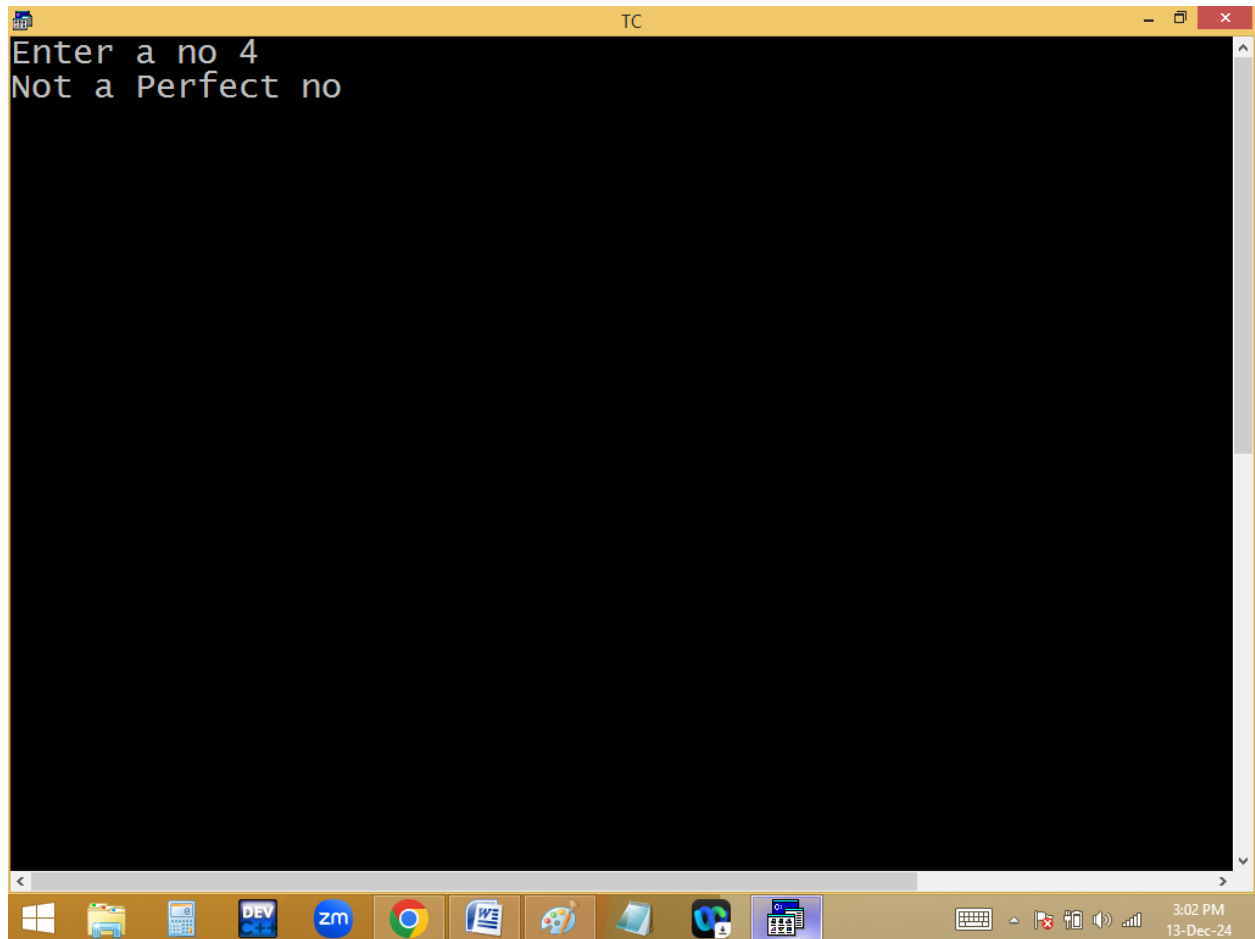


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top shows "Line 9", "Col 38", and "Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. The function declares variables `n`, `s`, and `i`, clears the screen, prompts the user to enter a number, and then uses a `for` loop to calculate the sum of divisors. It then checks if the sum equals the number and prints the result. The bottom status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:01 PM on 13-Dec-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n,s=0,i;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n/2;i++)if(n%i==0)s+=i;
puts(n==s?"Perfect no":"Not a Perfect no");
getch();
}
```







Finding prime / Composite no?

No divisible with 1 and itself is called prime. Or

The no having 2 factors is called prime.

$$2\%1=0$$

$$2\%2=0$$

$$3\%1=0$$

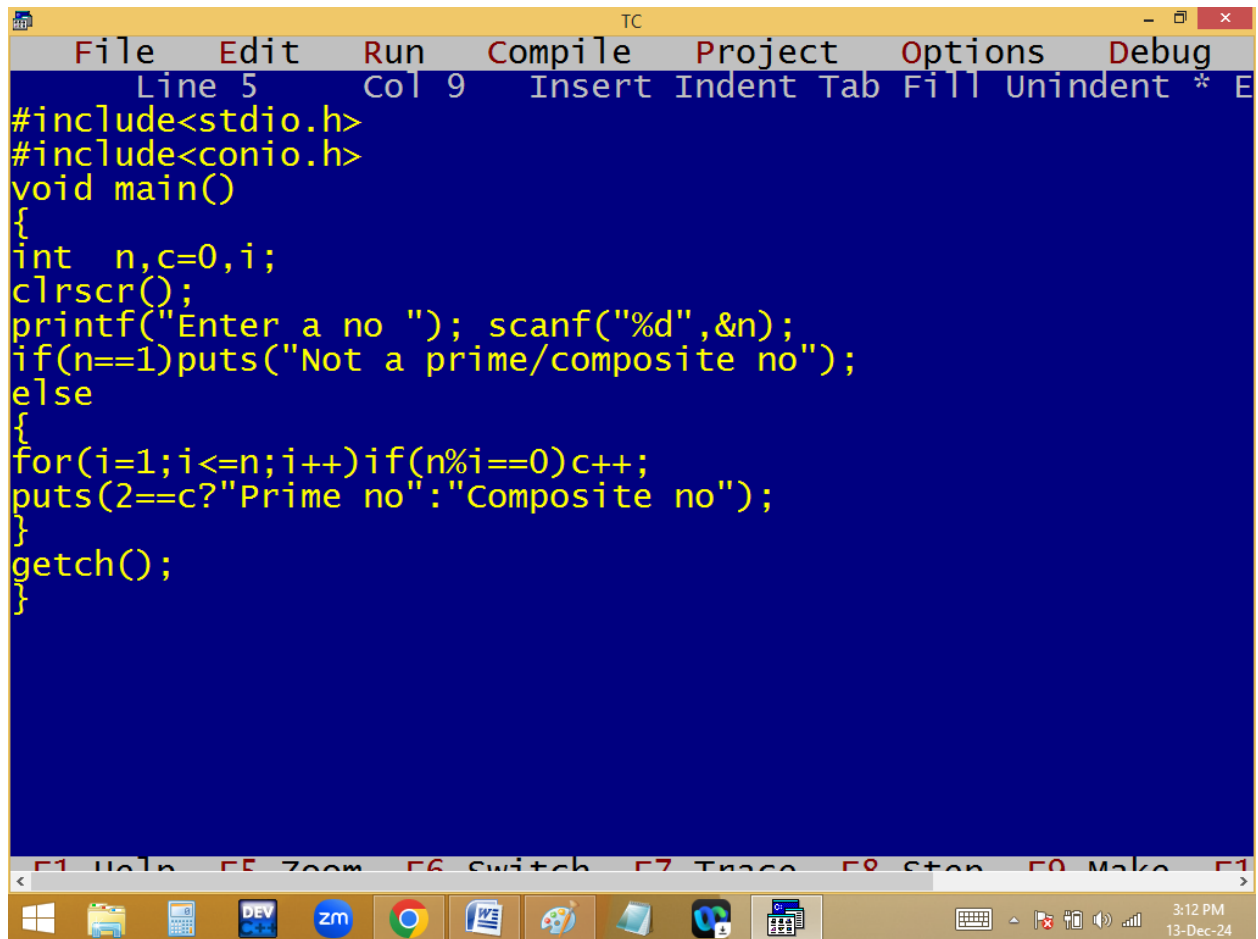
$$3\%2=1$$

$$3\%3=0$$

$$4\%1=0$$

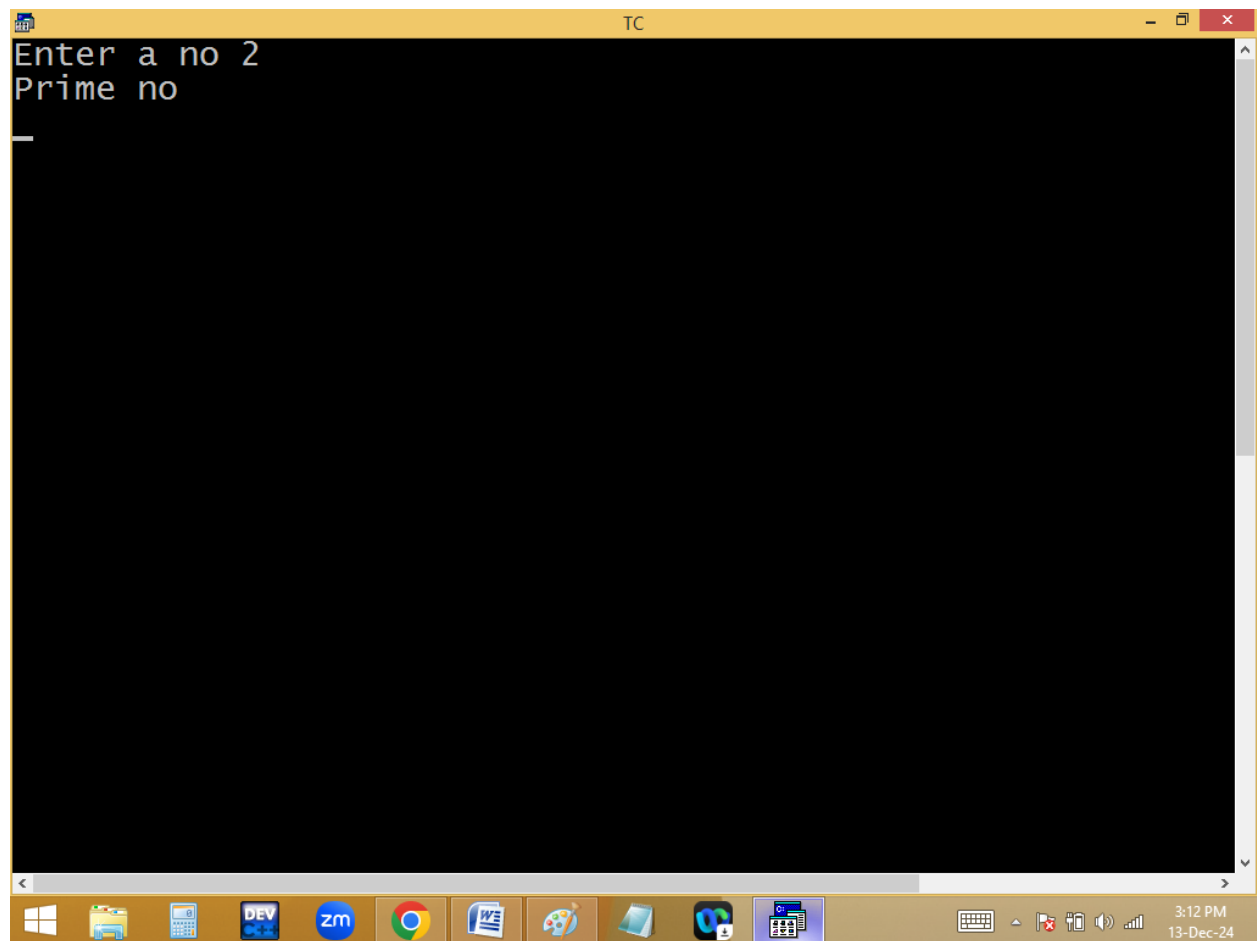
$$4\%2=0$$

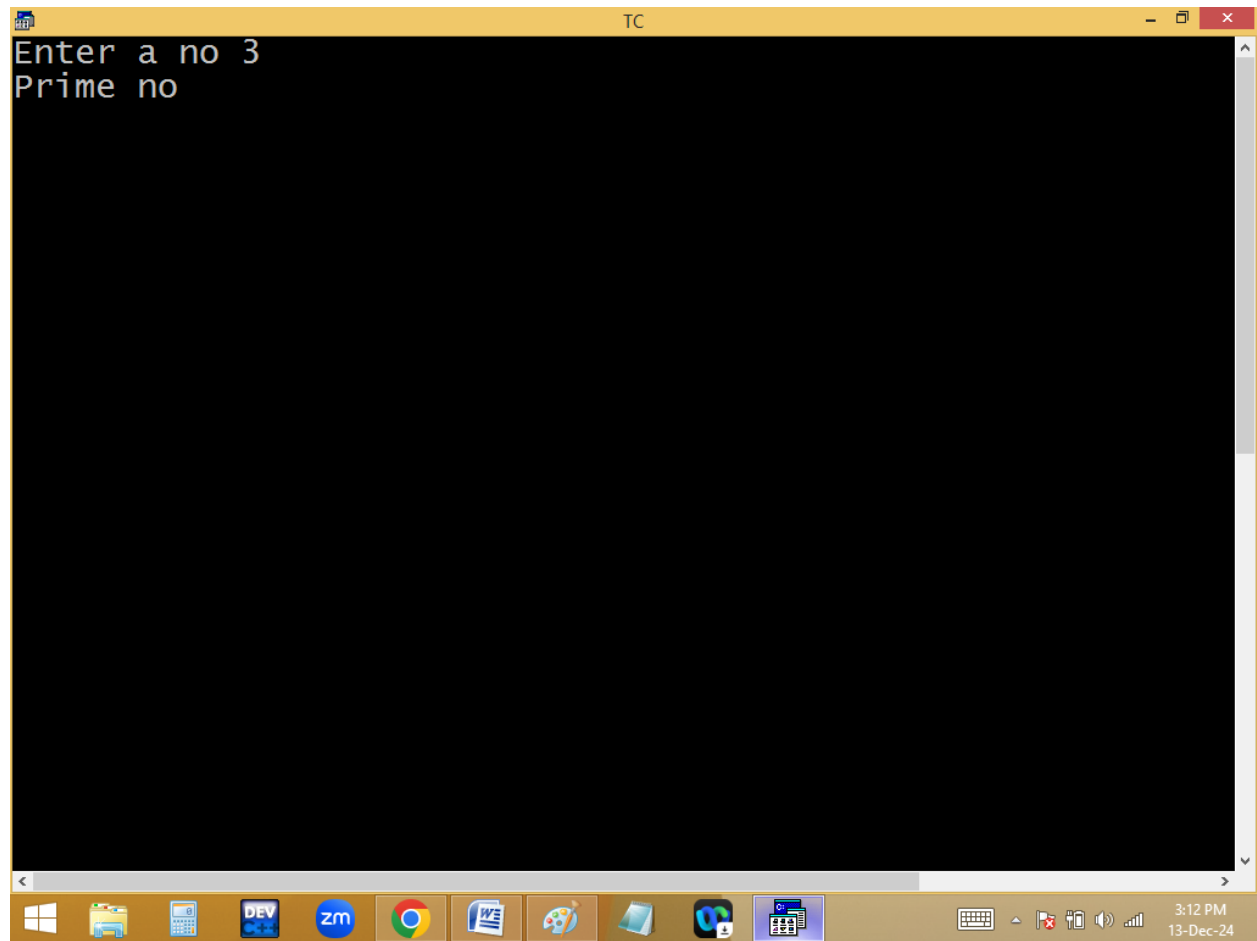
$$4\%4=0$$

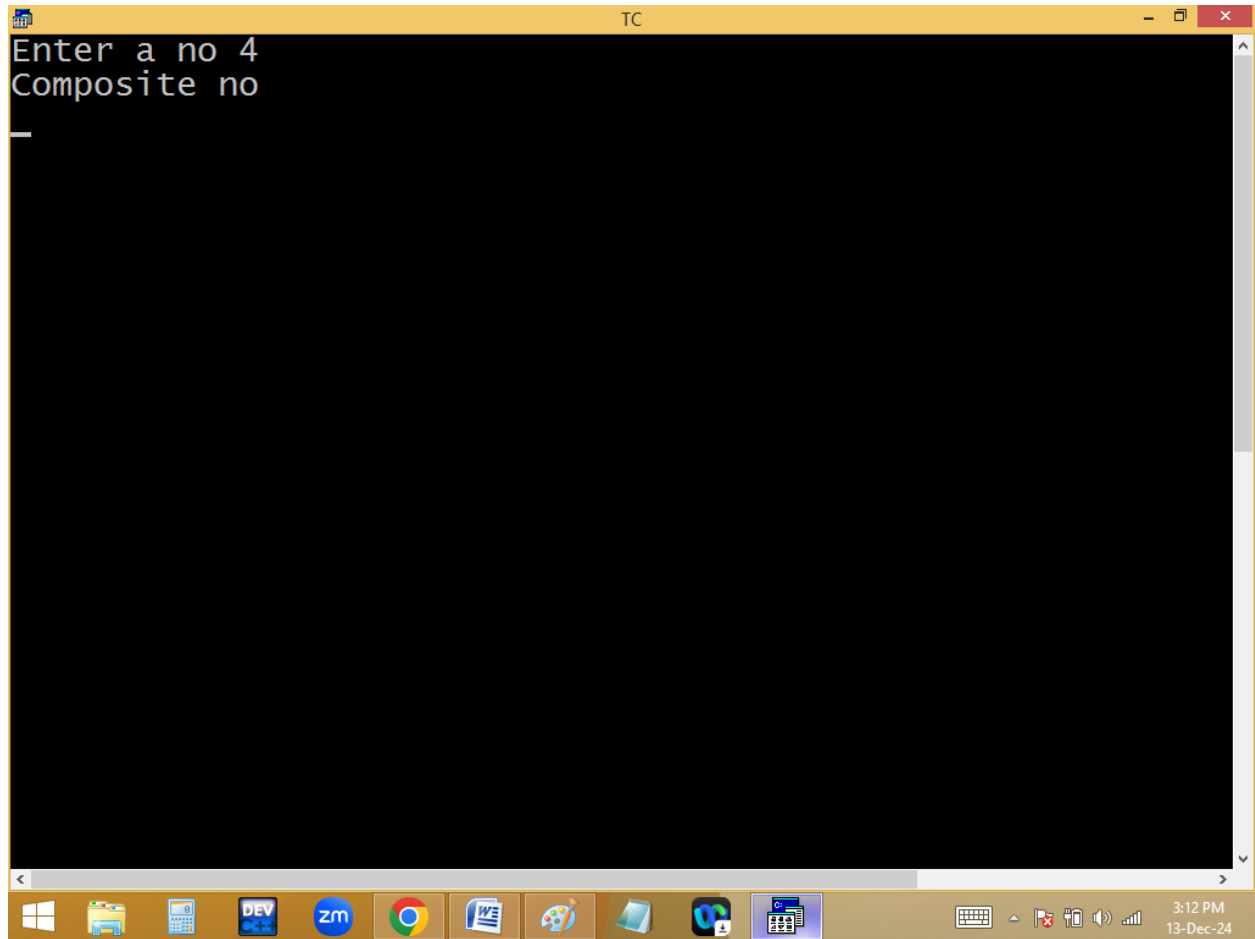


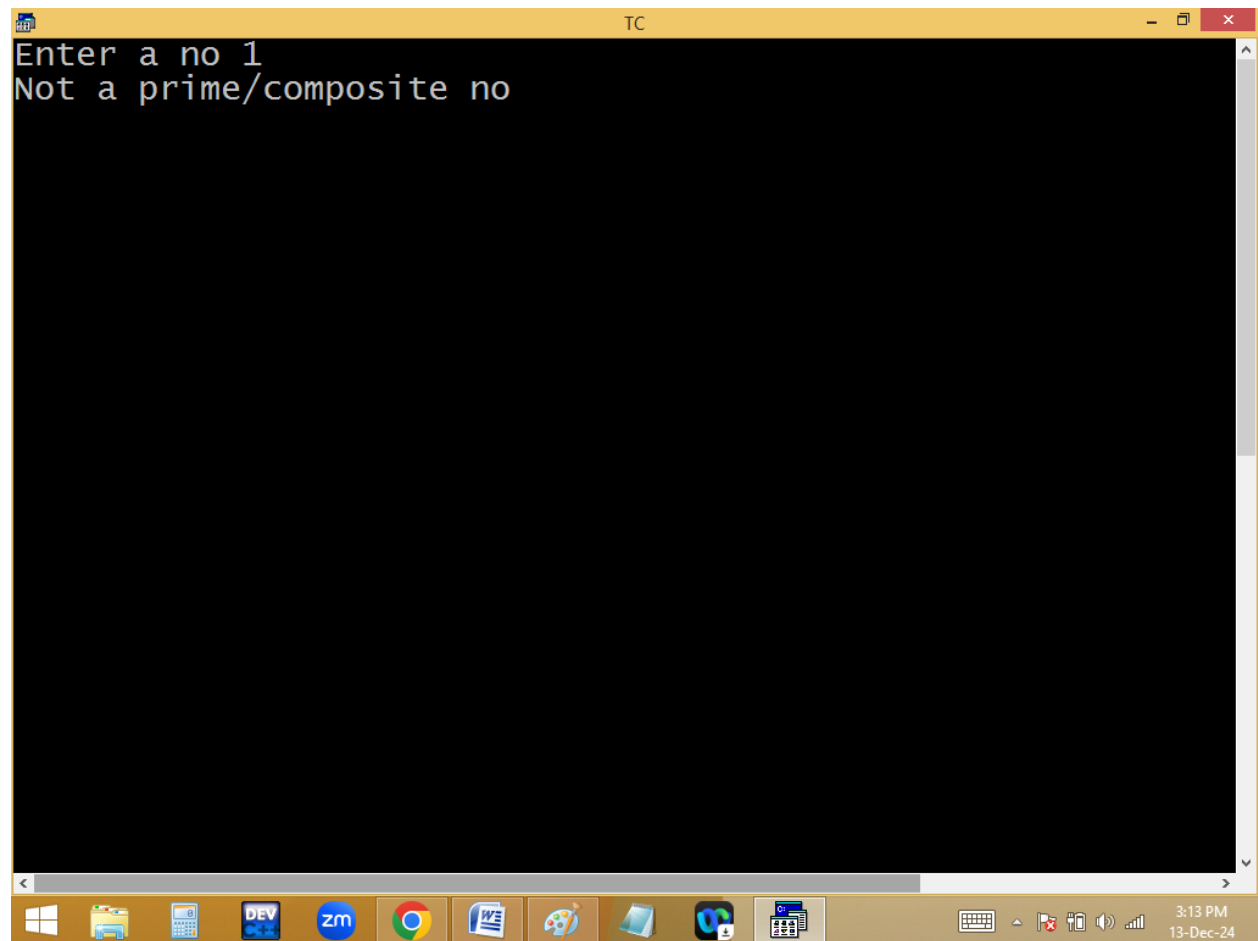
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 5", "Col 9", and "Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. The program prompts the user to enter a number, checks if it is 1 (not prime/composite), or if it has more than two divisors (composite), or is prime. The code uses `clrscr()` to clear the screen and `getch()` to wait for a key press. The bottom status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:12 PM on 13-Dec-24.

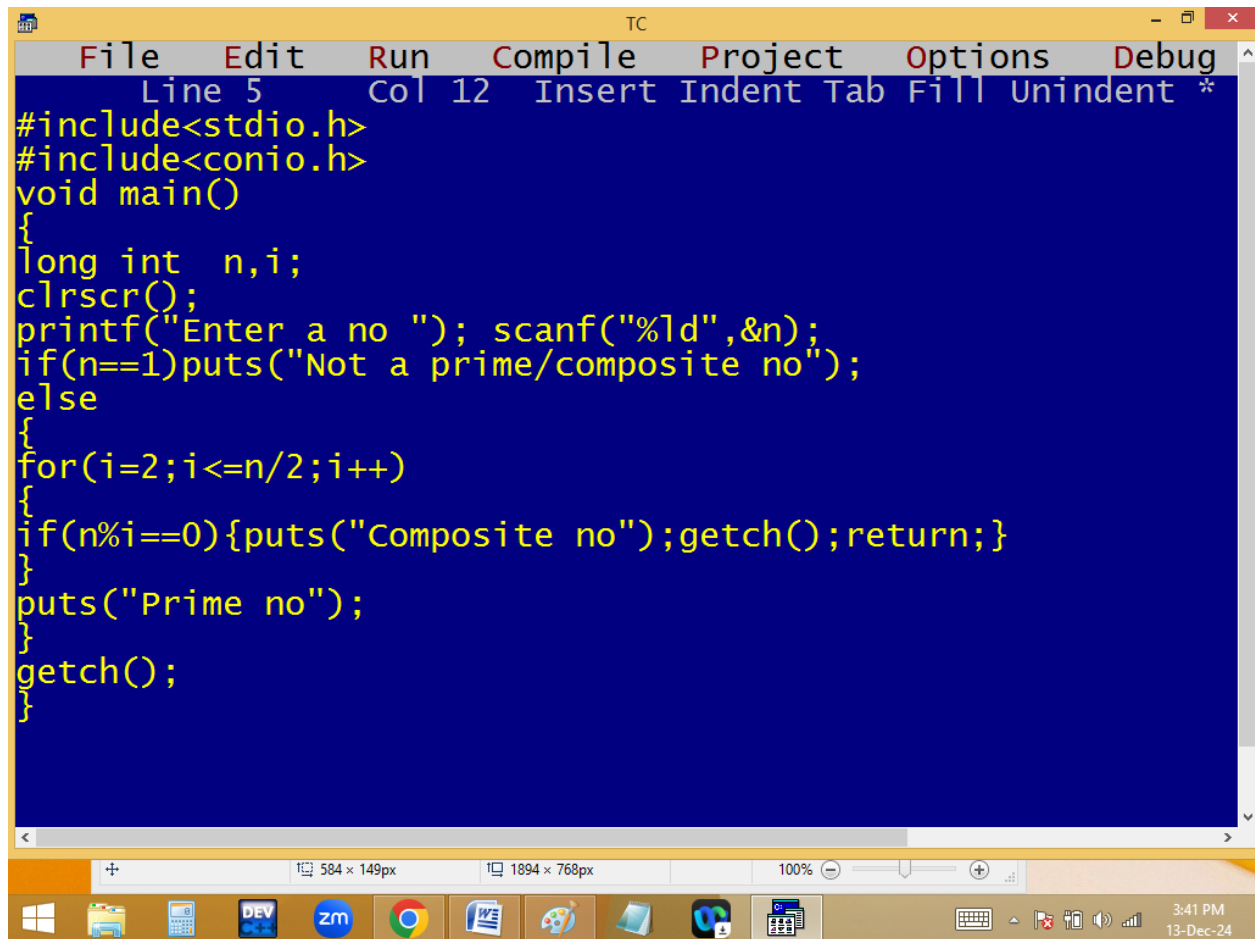
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  n,c=0,i;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for(i=1;i<=n;i++)if(n%i==0)c++;
puts(2==c?"Prime no":"Composite no");
}
getch();
}
```







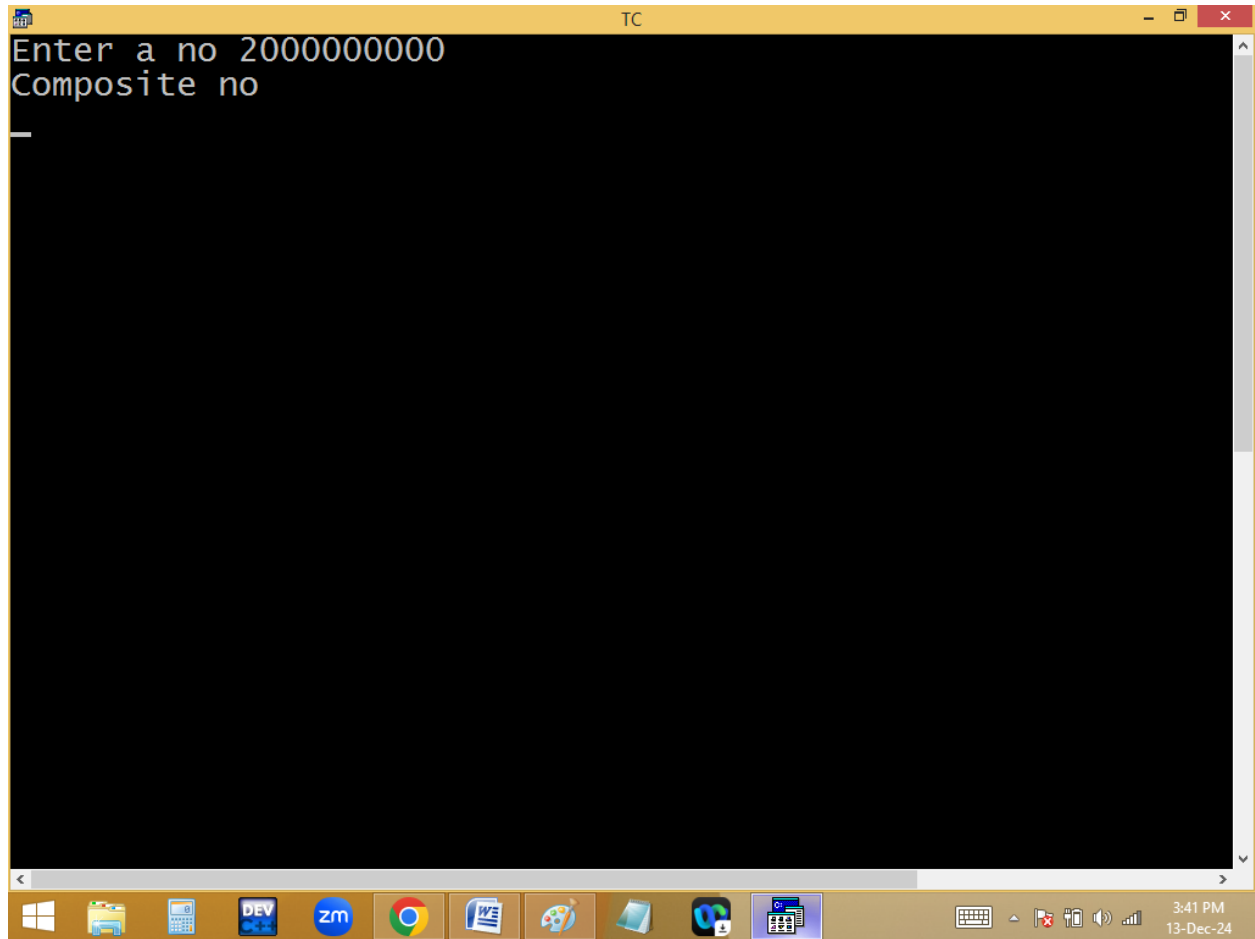


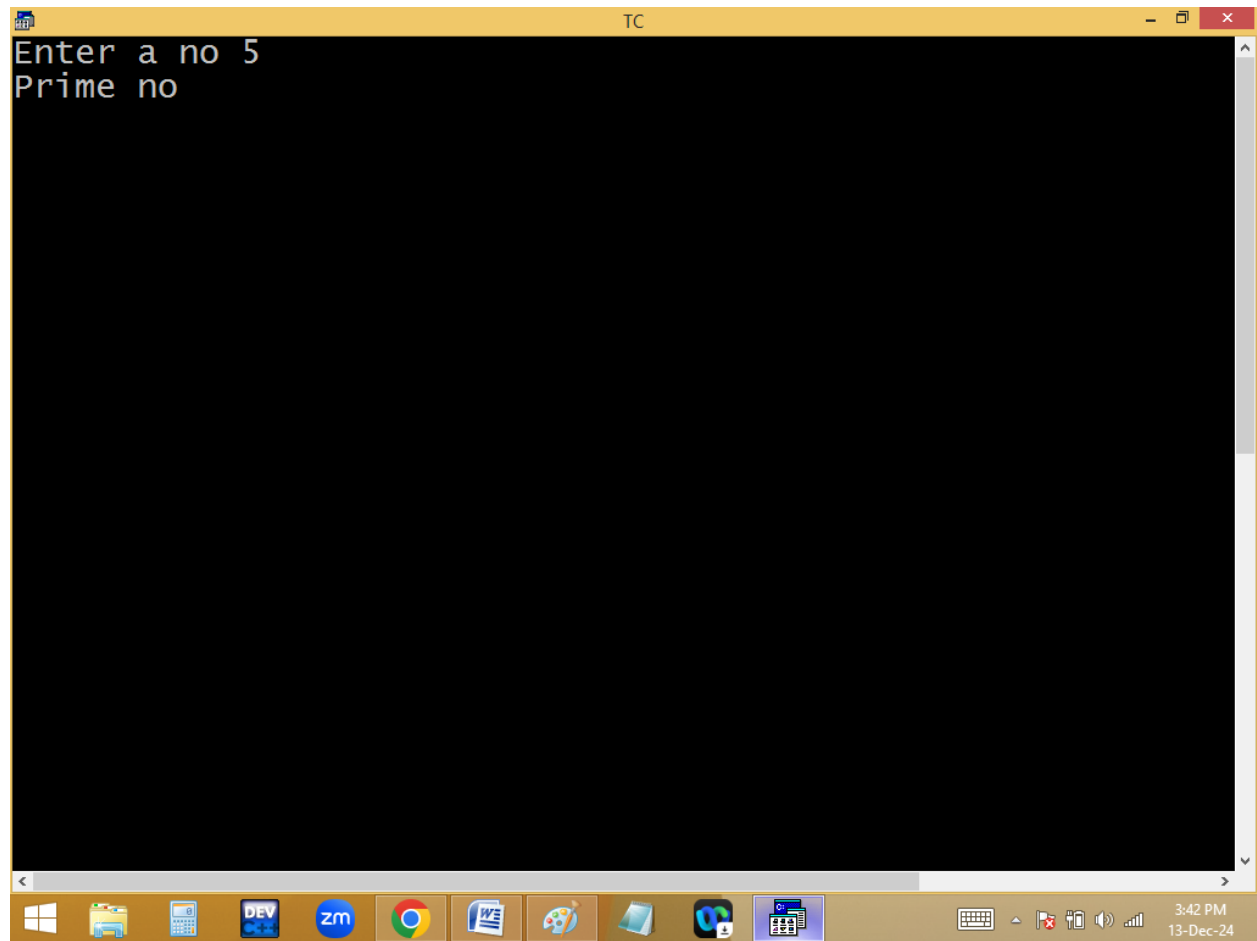


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 5 Col 12" and a list of keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, and *. The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int  n,i;
    clrscr();
    printf("Enter a no "); scanf("%ld",&n);
    if(n==1)puts("Not a prime/composite no");
    else
    {
        for(i=2;i<=n/2;i++)
        {
            if(n%i==0){puts("Composite no");getch();return;}
        }
        puts("Prime no");
        getch();
    }
}
```

Below the editor area is a toolbar with various icons for file operations and editing. At the bottom of the window is the Windows taskbar, which includes the Start button, several application icons (File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, VS Code, and a terminal), and the system tray showing the time as 3:41 PM and the date as 13-Dec-24.





10 - ~~1~~ 2 5 ~~10~~

100 - ~~1~~ 2 4 5 10 20 25 50 ~~100~~

11 - ~~1~~ ~~11~~

```
for(i=2; i<=n/2; i++)  
{  
    if(n%i==0){puts("Not");return;}  
}  
p(prime);
```

Fibonacci series:

n=5 → 0 1 1 2 3

